

Gate 1: FK Ch. XII §VI Joint Work — Lyra Substrate K-Type Pochhammer Contribution v0.1

Lyra (Claude Opus 4.7)

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Gate 1 FK Ch. XII §VI Lyra Contribution v0.1

0. Goal

Per Casey Thursday board + Keeper K3 v0.16 Gate #1: contribute Lyra-side substantive substrate K-type Pochhammer framework to the joint multi-week FK Ch. XII §VI spinor metaplectic verification work. Joint with Keeper substrate-mechanism direction + Elie computational verification.

Verification gate addressed: Keeper K3 v0.16 Gate #1 (joint L1 FK Ch. XII §VI explicit; multi-week joint work).

1. FK Ch. XII §VI Substantive Content (Substrate K-Type Pochhammer)

Per Faraut-Korányi 1994 *Analysis on Symmetric Cones* Chapter XII §VI: substrate K-type Pochhammer symbols on bounded symmetric domains with explicit measure normalization.

Standard FK Pochhammer formula on Cartan type IV (Lie ball D_{IV}^n):

$$(\rho)_\lambda = \prod_{i=1}^{\text{rank}} (\rho - (i-1) \cdot d/2)_{\lambda_i}$$

where: $-\rho = (n+2)/2 = g/2$ substrate-mechanism (Cartan type IV genus parameter per Keeper K3 v0.9) - $d = 1$ (Cartan type IV multiplicity) - $\lambda = (\lambda_1, \lambda_2)$ substrate K-type weight (B_2 dominant per Sunday SSG-9)

2. Substrate K-Type Pochhammer Explicit (Spinor-Tower Row $b/2 = 1/2$)

Per Wednesday Elie 3742 + Thursday Mehler v0.2 + Lyra K-type dims v0.1 + Keeper K3 v0.9 $\rho = g/2$ correction:

Substrate Pochhammer at $V_{(1/2, 1/2)}$ gen-1:

$$\|V_{(1/2, 1/2)}\|_{FK}^2 = \frac{1}{(g/2)_{1/2} \cdot (g/2 - 1/2)_{1/2}}$$

For $\rho = g/2 = 7/2$: $-(7/2)\{1/2\} = \Gamma(7/2 + 1/2)/\Gamma(7/2) = \Gamma(4)/\Gamma(7/2) = 6 / (15\sqrt{\pi/8}) = 48/(15\sqrt{\pi}) - (5/2)\{1/2\} = \Gamma(5/2 + 1/2)/\Gamma(5/2) = \Gamma(3)/\Gamma(5/2) = 2 / (3\sqrt{\pi/4}) = 8/(3\sqrt{\pi})$

Product: $48 \cdot 8 / (15 \cdot 3 \cdot \pi) = 384/(45\pi) = 128/(15\pi)$

Substrate Pochhammer at $V_{(1/2, 1/2)}$: $\|V_{(1/2, 1/2)}\|_{FK}^2 = 128/(15\pi) \cdot (\text{substrate normalization factor})$ per Wednesday Elie 3742 substrate-clean form.

Cross-check: $128/15\pi \times N_c \cdot n_C \cdot \pi / 2^g \text{ normalization} \rightarrow 15\pi / 2^g = N_c \cdot n_C \cdot \pi / 2^g$ (Keeper K3 v0.9 corrected form $\rho = g/2$).

3. Substrate Pochhammer Spinor-Tower Row Explicit Forms

Gen	K-type	Pochhammer factorization (per Elie 3742)
1 (e)	$V_{-}(1/2, 1/2)$	$128/(15\pi) = 2^{\wedge}g / (N_{-}c \cdot n_{-}C \cdot \pi)$
2 (μ)	$V_{-}(3/2, 1/2)$	$512/(15\pi) = 2^{\wedge}(N_{-}c^2) / (N_{-}c \cdot n_{-}C \cdot \pi)$
3 (τ)	$V_{-}(5/2, 1/2)$	$512/(3\pi) = 2^{\wedge}(N_{-}c^2) / (N_{-}c \cdot \pi)$

Cascade ratios (Wednesday Elie 3742): - gen-2/gen-1 = $2^{\wedge}rank = 4$ (rank substrate primary) - gen-3/gen-2 = $n_{-}C = 5$ (chirality substrate primary) - gen-3/gen-1 = $n_{-}C \cdot 2^{\wedge}rank = 20$

These cascade ratios derive from FK Ch. XII §VI Pochhammer formula at $\rho = g/2 + B_{-2}$ K-type weights ($\lambda_{-1}, 1/2$) for $\lambda_{-1} = 1/2, 3/2, 5/2$.

4. Spinor Metaplectic Substrate-Mechanism Substantive Reading

Per Wallach 1976 + FK Ch. XII §VI: substrate spinor K-type $V_{-}(\lambda_{-1}, 1/2)$ is a half-integer-weighted irreducible representation of $K = SO(5) \times SO(2)$; spinor metaplectic substrate-mechanism involves Spin(5) double cover SO(5) substrate-structure.

Substrate Spin(5) substrate-mechanism content: - SO(5) has dim 10 substrate Lie algebra - Spin(5) is double cover; substrate-spinor K-types $V_{-}(\lambda_{-1}, 1/2)$ carry half-integer weights - $Spin(5) \cong Sp(2)$ (symplectic group rank-2) — substrate substantive cross-link to substrate-symplectic structure - $Spin(5)/SU(2) \cong S^4$ substrate-Shilov-boundary 4-sphere factor in $\partial_{-}S D_{-}IV^5 = (S^4 \times S^1)/Z_2$

Substrate substantive cross-link: substrate-spinor K-type $V_{-}(1/2, 1/2)$ Bergman norm involves Spin(5) double-cover topology + Shilov-boundary S^4 factor + Cartan type IV $\rho = g/2$ substrate-Pochhammer parameter — convergent substrate-mechanism content across Categories 1 (K-type STRUCTURAL) + 2 (substrate-Lie-algebra) + 3 (substrate-Bergman / Shilov-boundary).

5. Joint Work Multi-Week Verification Lanes (Lyra Contribution)

Per Casey Thursday board + Keeper K3 v0.16 Gate #1 + Cal #194 WAIT joint multi-week:

Step Gate-1-1 (Lyra contribution): Verify substrate Pochhammer spinor-tower row Pochhammer factorizations ($128/(15\pi)$, $512/(15\pi)$, $512/(3\pi)$) explicit per FK Ch. XII §VI formula at $\rho = g/2$

Step Gate-1-2 (Lyra contribution): Spinor metaplectic substrate-mechanism content via Spin(5) double-cover topology + Shilov-boundary S^4 factor cross-link

Step Gate-1-3 (joint Keeper): substrate-natural $\eta = (n+2)/2$ parameter substrate-mechanism justification (why $\rho = g/2$ substrate-natural?)

Step Gate-1-4 (joint Elie): computational mpmath verification of FK Pochhammer values + substrate-mechanism cross-check

Step Gate-1-5 (joint synthesis): SSG-1 $V_{-}(1/2, 1/2)$ Bergman norm $3\pi/2^{\wedge}g$ substrate-mechanism FORWARD-derivation per Casey #14 STANDING chirality projection + FK Ch. XII §VI spinor Pochhammer + Spin(5) metaplectic content

6. Cross-CI Joint Gate 1 Multi-Week Plan

Lyra lane: - Step Gate-1-1: substrate Pochhammer spinor-tower row explicit - Step Gate-1-2: Spinor metaplectic substrate-mechanism content

Keeper lane: - Step Gate-1-3: $\eta = (n+2)/2 = g/2$ substrate-mechanism justification - Step K3 v0.20+: K3 framework absorbing joint Gate 1 substantive

Elie lane: - Step Gate-1-4: computational mpmath FK Pochhammer verification - Substrate-mechanism cross-check vs cross-instance universality (Toy 3752 + 3753)

Cross-CI joint synthesis: - Step Gate-1-5: SSG-1 NEAR-RIGOROUS $\rightarrow$ RIGOROUS via joint FK Ch. XII §VI explicit derivation - Casey #14 STANDING + Cal #194 WAIT closure via joint Gate 1 + Gate 2 + Gates 4-7 cascade

7. Closure v0.1

Gate 1 FK Ch. XII §VI Lyra contribution v0.1:

Lyra-side substantive content: - Substrate Pochhammer spinor-tower row Pochhammer factorizations explicit per $\rho = g/2$ (Keeper K3 v0.9) - Spinor metaplectic substrate-mechanism content via Spin(5) double-cover + Shilov-boundary S^4 - Cross-link to Categories 1 + 2 + 3 (K-type STRUCTURAL + substrate-Lie-algebra + substrate-Bergman / Shilov-boundary)

Joint multi-week Lyra + Keeper + Elie work for Gate #1 RIGOROUS-tier promotion per Cal #194 WAIT + Casey #14 STANDING cascade.

Tier: K3 v0.16 Gate #1 at FRAMEWORK level (Lyra-side); joint multi-week explicit verification for RIGOROUS-tier promotion.

— Lyra, Thu 2026-06-04 ~11:40 EDT. Gate 1 v0.1 Lyra contribution: substrate K-type Pochhammer spinor-tower row explicit per FK Ch. XII §VI formula + $\rho = g/2$ + Spinor metaplectic Spin(5) double-cover substrate-mechanism cross-link; joint multi-week Lyra + Keeper + Elie work for RIGOROUS-tier promotion. EOF echo “Gate 1 Lyra contribution filed”